CHINA TRANSFORMED

Historical Change and the Limits of European Experience

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ECONOMIC CHANGE IN LATE IMPERIAL CHINA AND EARLY MODERN EUROPE

The decades since the early 1960s have seen a wealth of research on Chinese economic history, principally by Chinese and Japanese researchers. Scholars have identified a cluster of major changes beginning in roughly the tenth century, especially in the area of eastern China near modern-day Shanghai known as Jiangnan in the late imperial period. Agricultural land productivity rose because of new seed varieties and technological improvements; cash cropping expanded as increased trade carried goods to growing cities and towns. These changes were supported by improved transportation, the formation of merchant organizations, and government policies that gave freer rein to market exchange.

A broad consensus also exists regarding the general status of China's contemporary economy—as its industries grow, China is only beginning to experience sustained economic growth with rising standards of living; only since the 1980s has the transformation of the agrarian basis of life shared by hundreds of millions of people begun in earnest. Between these two points of general agreement that bracket a millennium of Chinese history, we lack a fully persuasive interpretation of Chinese economic changes.[1]

[1]Feuerwerker (1992) identifies some of the analytical problems that confront economic historians of China. My approach to these problems complements his effort to distinguish among sources of growth as a strategy to explain more effectively economic change in Chinese history.

The Problem of Economic Change in Chinese History

Perhaps the central reason for the historians' failure to explain what happened to China's economy in mid- and late imperial times lies in the continued effort to frame the problem in terms of what *didn't* happen. China did not follow any European path of economic development. Studies of Chinese economic practices between 1500 and 1900 usually proceed in one of two ways. They pinpoint a Chinese difference from Europe and label this the key differentiating factor; for example, Nishijima's classic study (1966) of cotton handicrafts shows how production was tied to rural industry. Alternatively, research highlights a small cluster of similarities that promised European-style success for China but did not deliver. Studies of agricultural commercialization, the use of hired labor, the expansion of handicraft production, and urban factory formation, among others, often finger one of three villains to explain the absence of a more cheerful storyline (Liu Yongcheng 1982: Li, Wei, and Jing 1983; Wu and Xu 1985; Zhang Guohui 1986). The first is the Chinese state, which obstructs progress to protect its power. The second is imperialism, which warps economic changes to serve foreign interests. The third group of villains are "feudal" powerholders anxious to defend their elite positions against change. Some studies are framed with a mechanical Marxist perspective. But the attempt to define what went wrong in China according to what went right in Europe informs non-Marxist scholarship as well. Mark Elvin's influential 1973 study, *The Pattern of the Chinese Past,* asks why China did not experience European-style scientific and organizational changes after achieving a "medieval economic revolution." More recently, Philip Huang (1990) explores the divergence of Chinese experiences from European ideal types that the author associates with Smith and Marx. Nor is the search for what Chinese economic history lacks confined to China specialists. John A. Hall (1985:56) speaks of Chinese "institutional blockages to the market" in his wide-ranging assessment of world history from the perspective of European development, whereas E. L. Jones, in the second of his thought-provoking comparative economic histories (1988:141), suggests the Chinese state stifled development through "under-government," the standards for which are European government policies.

What unites almost all assessments of economic change in late-imperial China is an agreement that China possessed ingredients for continued economic change based on the dramatic developments of the Song period. The Marxist analysis looks for villains that blocked what would otherwise have been a natural set of developments toward capitalism. Though resolutely not Marxist, E. L. Jones (1981, 1988) offers a similar perspective on economic growth as a natural phenomenon that once on track continues to reproduce itself unless taken off track by some arbitrary interference. For China, he sees the government as the force that both fails to provide the necessary framework for continued growth and blocks what had been positive developments. He typifies a perspective that simultaneously makes the state too weak to have been positive but strong enough to have been negative.

Although most analysts search for positive obstructions to economic development in China, a smaller group searches for a key ingredient that the Chinese lacked. The most famous of such analyses remains Max Weber's contrast of the Protestant ethic with religious beliefs in other parts of the world. The argument that Protestantism was particularly conducive to the development of capitalism has been qualified in several ways. Within Europe, Catholic areas also achieved economic development. Beyond Europe, arguments have been made for the fit between religion and economic change. For China, Yu Ying-shih (1987) has shown how new concerns in sixteenth- and seventeenth-century Confucianism paralleled the rise of a distinctive merchant point of view as trade expanded in importance. For eighteenth-century Japan, Tetsuo Najita (1987) has shown how a group of Osaka merchants created a respected place for themselves in a Confucian world view. It is difficult to isolate a set of beliefs that proved crucial to social change for two reasons. First, the situations in China and Europe suggest that at least partially parallel kinds of intellectual change can appear in different societies without producing the same dynamics of economic change. Second, economic developments in Catholic and Protestants parts of Europe suggest that both areas in which religion changes and those where it does not can experience the same economic changes. The linkages between intellectual belief and economic change are simply too complex to reduce to simple expectations about the impact of religious belief on economic behavior.

Explaining what happened in Chinese economic history by appeal to what did not happen raises several difficulties. For one, it precludes the assessment of Chinese practices that do not fit neatly into European categories.[2] For another, it makes explanation too easy. The changes that led to European capitalism include numerous elements that were historically distinctive. One can find with little effort any number of differences between China and Europe, but assessing which of these differences mattered is difficult without developing analytical standards of significance.[3] One source of such standards is a baseline of similarities, which can then be used to delineate the arena within which important initial differences occurred. Other differences can then be introduced as we explore further the distinctive paths followed by different parts of Eurasia. Otherwise, all differences compete for our attention. The economic similarities considered here begin with Adam Smith.

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[2]Eurasian differences in state making and political economy appear to have been partially obscured by an insistent adoption of European standards of state making and political economy. This issue is explored further in Part II.

[3]For instance, if we attribute England's Industrial Revolution to its abundance of resources and skilled mechanics, an appropriate approach to technical problems, a supportive political system and social structure that developed property rights, patent laws, rational taxes, and a laissez-faire approach to economic activity, we end up with a broad-based description rather than an explanation of the Industrial Revolution (Snooks 1994: 14–15).

Early Modern European Dynamics of Growth

The driving force behind economic improvements in Adam Smith's *Wealth of Nations* is productivity gains attending division of labor and specialization. By producing what they are best suited to produce and exchanging their products with others, people capture the benefits of comparative advantage at the market place. Division of labor is limited only by the extent of the market. As the market expands, the opportunities for Smithian growth increase accordingly. A decentralized price system widens the scope of the market and extends the advantages accruing from the division of labor (Blaug 1985:61). These dynamics of expansion were qualified in early modern Europe by the rhythm of demographic change and unpredictable fluctuations in harvests.

The demographic losses associated with the Black Death of 1348–50 decimated populations from the Black Sea through the Mediterranean to northern Europe.[4] Cities and towns were especially hard hit, with a cluster of economic consequences. First, handicraft production was disrupted, and trade between cities declined. Second, urban demand for agricultural goods fell, and farmers shifted from agriculture to livestock rearing in many areas. The gradual demographic and economic recovery of Western Europe from the Black Death led in the sixteenth century to higher levels of total population and aggregate economic output. During the fifteenth and sixteenth centuries a new financial infrastructure emerged to support long-distance trade. The elaboration of banking and marketing institutions made possible increasingly sophisticated patterns of exchange that recognized division of labor and specialization in production. But these new developments rested upon the fragile economic base of agriculture. Harvest conditions determined the annual fluctuations in food prices, which in turn heavily influenced the labor costs of manufacturing. When successive poor harvests lowered real wages, nonagricultural production usually fell, so that harvest shortfalls triggered cyclical declines in industry as well as agriculture. This cycle, made famous by Ernest Labrousse (1990), marked the rhythm of the pre-nineteenth-century European economy through its longer phases of growth and decline.

Amidst the constant fluctuations that marked the gradual recovery and then growth of Europe's economy, significant shifts occurred in the continent's most active economic centers. As new marketing networks developed and changes in textile production and other crafts took place, older centers in the Mediterranean were replaced by northern centers, especially in Holland and England. Economic growth looks especially impressive, therefore, if one focuses only on the regions of greatest growth. If, however, we look at a larger Europe and recognize that some areas declined as others rose, while yet others remained relatively unaffected, economic growth in early modern Europe appears more modest. The success of Venice and Genoa came in part at the expense of Muslim traders. Italian traders were in turn overshadowed by the expansion of Dutch, then English, finance and trade across the Atlantic and into Asia. Since agricultural production bulked so large in overall economic activity, economies had little ability to grow much faster than the rates of expansion in agriculture where growth was the product of some combination of the following factors: (1) extended arable; (2) increased inputs of capital and labor; (3) increased specialization; and (4) technological improvements. Each of these factors played a role in expanding economic production. They were also related: for instance, increased capital went to technological improvements, and more labor was expended in response to market opportunities for specialization. Both the economy and population grew in the early modern period, but not without cycles of expansion and contraction. The seventeenth-century "general crisis" of economic, social, and political difficulties has included in some scholarly assessments a crisis of population and resources, a Malthusian crisis.

During the eighteenth century many parts of Western Europe increasingly participated in commerce through their agricultural and industrial production. England had broken free in the eighteenth century from the dire threat of famines, while in France subsistence crises were no longer of the killing magnitudes common a century before (A. Appleby 1969). The Europe Smith analyzed was certainly better off than it had been before, but it had yet to begin its nineteenth-century urban factory industrialization which led to a fundamental transformation of society and economy. The economy of Smith's *Wealth of Nations* remained principally an agricultural economy. No wonder then that Smith stressed agricultural investment, assumed economic growth was finite, and expected real wages ultimately to fall to subsistence levels (Caton 1985, Smith 1937, Blaug 1985:35–66). Malthus and Smith lived in the same world of limited economic possibilities. Smith's world was not Europe of the nineteenth century. In key ways, eighteenth-century Europe shared more with China of the same period than it did with the Europe of the nineteenth and twentieth centuries.

Smithian Dynamics in China

We witness Smithian dynamics across much of China between the sixteenth and nineteenth centuries. The broad features of increased cash cropping, handicrafts, and trade are well known in the Chinese and Japanese literature, even if Smithian propositions are not explicitly identified to explain parts of the process. Most famous are the expanding cotton and silk industries of the Lower Yangzi region near Shanghai, the two principal handicraft industries that joined rice and other cash crops to create China's richest regional economy. To feed the population of this area, rice grown in the upstream provinces of Anhui, Jiangxi, Hubei, and especially Hunan and Sichuan moved down the Yangzi River. Other cash crops and handicrafts, such as cotton, indigo, tobacco, pottery, and paper, emerged in parts of these provinces as market expansion connected an increasing number of locales (Wu and Xu 1985:82–95, 143–55, 272–76).

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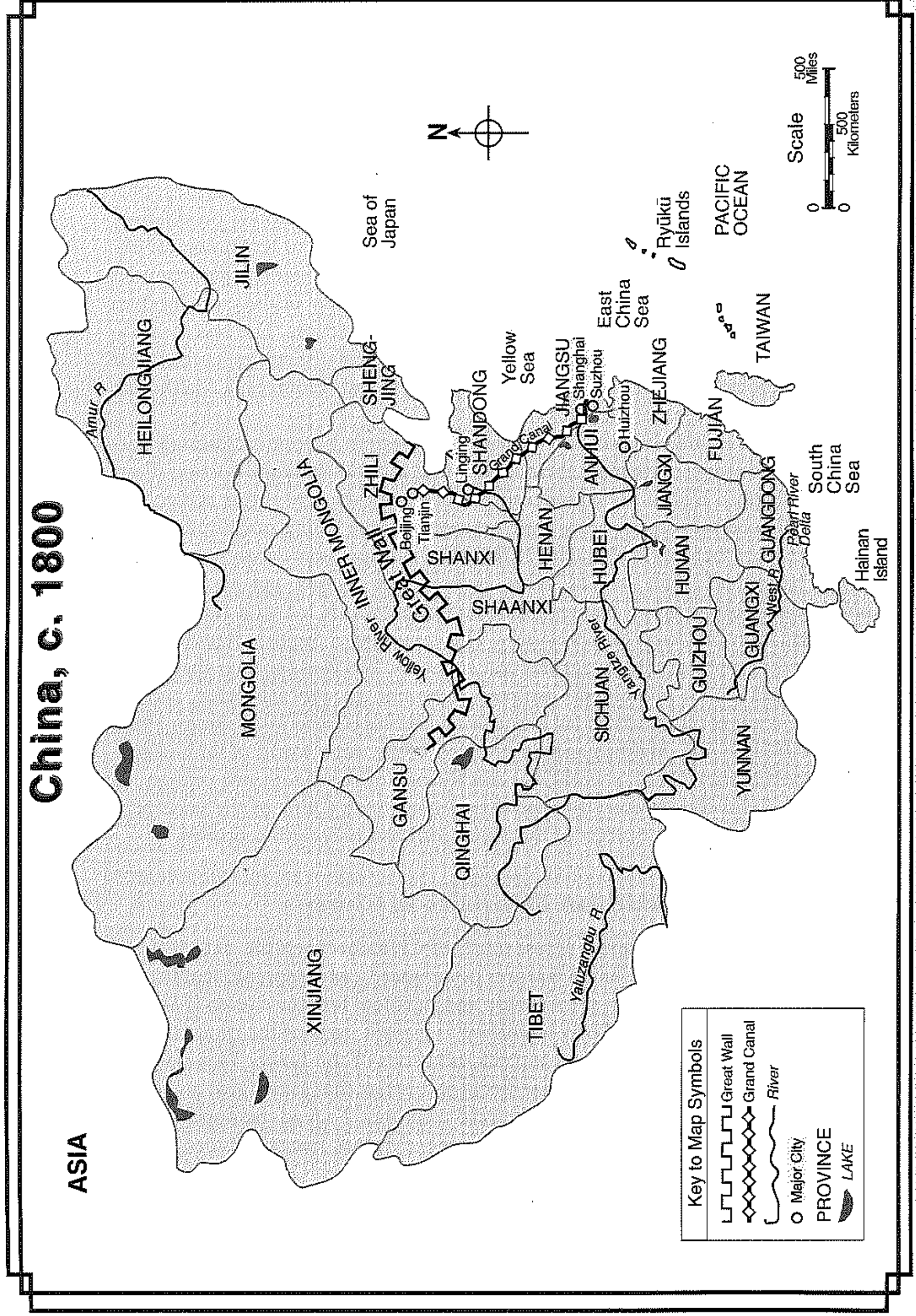
[4]The following paragraphs highlight some of the main themes in the economic history of this period. Among the sources consulted are Cipolla 1972–74, 1980; Lopez 1971; Miskimin 1969, 1977.



Market expansion was most salient along the Yangzi River, but hardly limited to this vast area. In south and southeast China, cash crops and handicrafts expanded in several areas. The Pearl River delta in Guangdong produced sugar cane, fruits, silk, cotton, ironware, and oils from sesame and tung plants (Tang and Li 1985; Luo Yixing 1985). Along the southeast coast, sixteenth-century foreign trade ties stimulated cash-crop production in tea and sugar (E. Rawski 1972). The northern half of the empire witnessed less market expansion, in large part because river transportation was more limited. But cash cropping, handicrafts, and trade did develop. Grand Canal towns like Linqing became major commercial centers with merchants who sold cloth, grain, pottery, paper, leather, tea, and salt (Xu Tan 1986). The hinterland of the city Tianjin became a center for fishing and salt, while Shandong province more generally witnessed market development led by the cash crops of cotton and tobacco (Guo 1989; Li Hua 1986).

Commercialization stimulated productivity increases. Studies of land productivity for both the lower Yangzi region and other parts of the Chinese empire show an increase over time after the tenth century. Improvements and extension of irrigation techniques, seed varieties, and farming methods have been documented. In general, land productivity increased with the application of additional fertilizer, the development of weather-resistant seed varieties well suited to local conditions, and more effective cultivation techniques (Li Bozhong 1984a, forthcoming; Min Zongdian 1984; Huang Miantang 1990). Labor productivity in agriculture rose in some cases, but whether there was much sustained increase in per capita income is far from clear. Growth may have been extensive much of the time, leading to an increase in total production, rather than intensive, with an increase in per capita productivity.

Working against the increased benefits derived from spatial divisions of labor and specialized production, a portion of the growing Chinese population in the eighteenth and nineteenth centuries turned to less fertile fields and to marginal economic pursuits. The clearance of hill lands during this period often brought into cultivation inferior food crops in conjunction with some cash crops and handicrafts. The case of southern Shaanxi, in northwest China, is instructive. This area had been the scene of some fighting in the late Ming dynasty (1368–1643); when the Qing dynasty (1644–1911) was established, migration repopulated the area and soon made it grow beyond its Ming population levels. New food crops on hill lands, including corn and sweet potatoes, complement wheat and millet to feed a population that included individuals in the timber trade, paper making, and iron workers (Fang 1979; Tan 1986; Xiao 1988; Chen and Zuo 1988). Development in this rather remote and none too fertile land did nevertheless take place. Market exchange was basic to the successes enjoyed by peasants. But since the basic resource possibilities of this part of China fell short of those in richer ecologies like the Yangzi River delta, labor productivities and standards of living could not match those of more favored regions. The same logic broadly applies to the many other hill land areas of China where land was opened during the Qing dynasty (Fu 1982; Zhang Jianmin 1987).[5]

The development of product markets provides the clearest indications of commercialization in China between 1500 and 1900. The increase in markets created a dense network of exchange. In Jiangnan the markets in some prefectures doubled or tripled in number between the late sixteenth and eighteenth centuries. Product specialization among markets also emerged with the rice trade centered at Suzhou and cotton and silk concentrated at other market sites (Liu Shiji 1987; Fan 1990). In addition to product markets, factor markets also emerged in some areas. For Jiangnan and southeast China, there is considerable evidence of a land market for both rentals and sales, which gave the seller the opportunity to repay the purchase price with interest during a specified number of years. If he failed to do so, he usually became the tenant of the buyer. These arrangements would persist over generations, with a son or grandson attempting to redeem land sold decades earlier (Yang Guozhen 1988).

Labor markets also developed for both short-term and long-term labor. Short-term labor filled two distinct purposes. It was important during the busy agricultural season, especially at harvest time. Conversely, during the winter slack season, peasants could search for off-farm employment. Other people accepted year-long contracts to work in agriculture. During the eighteenth century the inferior legal status of hired laborers tied to their positions as dependents within a household was gradually transformed; hired laborers became legally independent, if economically dependent upon others (Liu Yongcheng 1982:54–72; Li, Wei, and Jing 1983:243–516).

Analysis of China's late imperial credit markets lags behind studies of land and labor markets, not to mention product markets. Critiques of twentieth-century usury have led many to assume that the few late imperial credit markets were geared simply to consumption needs. But peasants in areas where trade was increasing turned to crops and crafts requiring capital to begin production and more modest annual infusions of capital to continue production. As Pan Ming-te (1994) has shown, the late imperial expansion of both cotton and silk textile production in Jiangnan depended on the availability of credit.

The combination of different factor markets meant that many peasant families made multiple adjustments to maximize production. In an ideal scenario, a peasant family could adjust their land by renting in or out and by selling or buying; they could adjust labor through hiring in or out; and they could seek loans to cover capital requirements. The absence of one or more of these markets limited the options available to a peasant. But wherever some of these factors markets existed, peasants had greater flexibility to use their resources efficiently to produce for the market than did peasants without markets.

[5]This does not mean, of course, that all migration was to less fertile areas. The repopulation of Sichuan and the opening of Manchuria stand out as the most salient instances of migration to fertile regions. For an overview of Qing migration movements, see Lee and Wong 1991.

Commercialization stimulated peasant efforts to combine land, labor, and capital to yield the greatest benefits. The pursuit of increased incomes did not mean Chinese peasants were any less liable to suffer poor harvests and even subsistence crises than were European peasants of the same period. As in Europe, there were considerable variations in the productivity and poverty of local agrarian economies. Regional economies in China, the equivalent in many cases to national economies in Europe, experienced cycles of expansion. Sixteenth-century expansion was most salient in the lower Yangzi, southeast coast, and South China regions. New groups of merchants organized expanded patterns of exchange that connected China's major cities to one another and to networks of market towns and the countryside around each of them.[6] After the economic decline initiated by the Ming rebellions followed by the disruptions caused by the Manchu invasion, the eighteenth century witnessed the resettlement of deserted land, the opening of new fields, and a renewed commercial expansion spanning even larger portions of the empire. Economic growth in the middle and upper Yangzi regions complemented growth in the lower Yangzi. Parts of North and Northwest China also increased production. The dynamics of Smithian expansion were present throughout.[7]

What then of economic change in the nineteenth century? With fundamentally different interpretations, to be sure, most Chinese and Western analysts of China's economy between 1850 and 1950 assign foreigners a crucial role. For Chinese Marxists, imperialism twisted and distorted China's path of development and blocked the country from replicating the stages of success enjoyed by Europe (Yan 1989). For Western-trained scholars, foreigners created the opportunities and offered the skills and techniques to build a modern economy in China (Hou 1965; Dernberger 1975). What both positions usually obscure is the degree to which the late nineteenth and early twentieth-century economic impact of foreigners was felt by most Chinese in terms of trade opportunities, the principles of which did not differ from those available in previous centuries. New commercial opportunities expanded the spatial scale on which Smithian dynamics worked; they did not fundamentally alter those dynamics. The regional impact of foreign markets was felt most strongly in the areas near transport nodes and along transportation routes. These included the recently opened treaty ports, the cities in which foreigners were allowed to reside and do business, and the new railroad lines that were laid in areas without decent river transport.

Railroads especially stimulated commercialization in northern areas that enjoyed close proximity to the lines; these areas began planting tobacco, peanuts, sesame, and soybeans. Foreign demand for tung oil spurred production in Yangzi

[6]The classic works on the two most important merchant groups are Terada 1972 on the Shanxi merchants and Fuji 1953–54 on the Huizhou (or Xin'an) merchants. On merchants groups more generally, see Fu 1956 and Zhang and Zhang 1993.

[7]The Chinese article and journal literature on Ming Qing economic expansion is voluminous. Two good book-length studies are Chen Xuewen 1989 and Zheng 1989.

River provinces and in the south (Liu Kexiang 1988). Per capita incomes in China's most prosperous farming regions which had the best access to international markets may well have risen as a result of international trade.[8] At the same time, there were surely instances where market changes created hardships like those of the early 1930s depression. But despite the positive impacts of international trade and the hardships that markets could cause, larger questions remain. How far could Smithian growth stimulate a modern industrialized economy in early twentieth-century China? Was a Malthusian crisis lurking in the background to threaten economic development?

Population Dynamics: Birth and Death Rates

Adam Smith stressed the limits on Chinese economic growth when he suggested that the country may have reached its height before Marco Polo's arrival. "It [China] had perhaps, even long before his [Polo's] time, acquired that full complement of riches which the nature of its laws and institutions permits it to acquire" (1937:71). Smith shared with Ricardo and Malthus a belief that economic growth itself was limited. All three asserted that subsistence costs and wages were dynamically linked through the economic determination of demographic rates. In Smith's estimation, high wages promoted the survival of children, and as more children survived, population growth drive wages down (1937:64–86). He also held that profit levels and interest rates fall in wealthy countries where opportunities to multiply riches have been exhausted (1937:87–98). Working within the same basic framework, Ricardo anticipated the exhaustion of natural resources (Blaug 1985:88), while Malthus feared the multiplication of populations beyond what their resource bases could support.

Though we conventionally associate Smith with the study of modern economic development, he and other classical economists interpreted a world yet to experience the massive industrial changes of the nineteenth century. As Mark Blaug counsels, "We need to remember that when the book [*The Wealth of Nations*] appeared, the typical water-driven factory held 300–400 workers, and there were only twenty or thirty such establishments in the whole of the British Isles. This helps to account for Adam Smith's neglect of fixed capital and for the convention, which he never really abandoned, that agriculture and not manufacture was the principal source of Britain's wealth" (1985:37). Smith, Ricardo, and Malthus all lived in a world in which agriculture remained the dominant sector of the economy.

The conventional picture of China poised on the brink of Malthusian crisis due to population growth outstripping resources is a powerful one. Its persuasiveness builds upon early twentieth-century perspectives of observers from industrialized countries and more recently upon contemporary perspectives that identify massive populations as a drag on modern economic growth in developing countries. A comparative analysis of China and Europe in early modern times qualifies these perspectives: putative differences in Chinese and European demographic histories are less certain than usually assumed. First, it should be noted that the broad rhythms and rates of population expansion were similar in China and Europe between 1400 and 1800 (Snook 1994:18). In general, the economies of both China and Europe expanded to support growing populations. Certainly there were fluctuations in living standards over time and regional differences as well, but estimates of aggregate population in China and Europe do not reveal fundamentally different population and resource conditions in China and Europe before 1800.

The crux of the matter concerns birth and death rates. In northwest Europe, where preventive checks of late marriage and a relatively large proportion of never-married women kept fertility rates below their biological maxima, high mortality rates were not necessary to hold population growth in check. China is seen as a land of high fertility and high mortality, population stability being achieved through the positive check of high mortality. An examination of the evidence on fertility and mortality does not, however, support this stereotypical contrast between China and Europe.

The only systematic analyses of late imperial fertility in fact show two types of preventive check, although these differ from the types encountered in Europe. Between 1700 and 1840, first births among the Qing nobility came approximately thirty-seven months after marriage, roughly double the interval among European populations. At the other end of the reproductive span, women stopped bearing children at about age 34, in contrast to European women, who did not stop until age 40 (Wang, Lee, Campbell 1995). As a result, China's marital fertility was much lower than people previously assumed.

Similar results were obtained in research on a rural population in northeast China for the century beginning in 1774, where fertility rates were only two-thirds those of comparable European populations. Again, late onset and early cessation of childbearing proved important. In addition, breast feeding appears to lengthen the period of infecundity and hence to increase birth intervals and reduce the total number of children born (Lee and Campbell 1957:93–94). While longer birth intervals due to breast feeding are found in numerous settings, the Chinese preventive checks of late starting and early stopping contrast sharply with European practices.[9] Though the particular preventive checks differed in China

[8] Loren Brandt unveils some dramatic estimates of growing per capita incomes and rising labor productivity in provinces along the Yangzi River, which he attributes in large measure to the integration of Chinese markets with world commodity markets. Through a variety of indirect evidence Brandt estimates an increase in per capita incomes of 44 percent and a rise in labor productivity in agriculture of 40 percent between the 1890s and 1930s (1989:133). Doubts about the data, estimation techniques, and assumptions Brandt uses suggest we not place too much faith in these conclusions. See R. B. Wong 1992 for a fuller discussion.

[9] An explanation for these differences would take us into the realm of cultural norms and social expectations about marriage and family. Briefly, it appears that the husband–wife relationship was less central to family life in China than in Europe. Regulation of conjugal relations was more possible and desirable in Chinese settings than in European ones.

and Europe, their effects were similar. Both tended to reduce rates of possible population growth.

The mortality component of demographic change is more complex. Certainly China, like Europe, suffered war, famine, and disease, the factors that Malthus considered positive checks on population growth. But China had another kind of positive check, one less often encountered in Europe and the result of deliberate parental action: infanticide. Long birth intervals and skewed sex ratios reflect the practice of female infanticide (Lee and Campbell 1997:65–70; Lee, Wang, and Campbell 1994). A reduction in the supply of women in turn leads to slower population growth in the next generation as the number of marriageable women falls. Together with the late onset and early cessation of births, the Chinese custom of infanticide acted as a sufficiently strong positive check to explain the generally similar patterns of population growth in China and Europe without invoking the positive check of high levels of adult mortality.

Other factors that affect mortality rates are less easily determined. It is difficult, for example, to prove that there is a direct link between mortality levels and availability of resources. Few people die of starvation except in crisis conditions. Many may suffer chronic malnutrition, but the effects of this condition on mortality are unclear, even among contemporary populations (Carmichael 1985, Livi-Bacci 1985, Scrimshaw 1985, Taylor 1985). Still, modern scholars continue to associate food supply conditions with levels of mortality (Simon 1985:218).

The idea that the Chinese suffered a low living standard that exposed them to higher mortality risks is a venerable one. Both Adam Smith and Thomas Malthus deplored the dietary conditions about which they had read in accounts of Chinese society. As Smith said, "The poverty of the lower ranks of people in China far surpasses that of the most beggarly nations in Europe. . . . The subsistence which they find . . . is so scanty that they are eager to fish up the nastiest garbage thrown overboard from any European ship. Any carrion, the carcase of a dead dog or cat, for example, half putrid and stinking, is as welcome to them as the most wholesome food to the people of other countries" (1937:72).

Malthus's assessment echoes Smith's: "If the accounts we have of it [China] are to be trusted, the lower classes of people are in the habit of living almost upon the smallest possible quantity of food, and are glad to get any putrid offals that European laborers would rather starve than eat" (1976:53).

Yet other European writers, including careful first-hand observers, came away with quite different assessments. Witness, for instance, Robert Fortune, a Scotsman who did not hold a very high opinion of Chinese agriculture but nevertheless noted: "for a few cash . . . a Chinese can dine in sumptuous manner upon his rice, fish, vegetables and tea; and I fully believe, that in no country in the world is there less real misery and want than in China" (1847:121, cited by Anderson 1988:96). In another volume he wrote, regarding the diet of tea-picking laborers:

> The food of these people is of the simplest kind—namely rice and vegetables, and a small portion of animal food, such as fish or pork. But the poorest classes in China seem to understand the art of preparing their food much better than the same classes at home. With the simple substances I have named, the Chinese labourer contrives to make a number of savoury dishes, upon which he breakfasts or dines most sumptuously. In Scotland, in former days—and I suppose it is much the same now—the harvest labourer's breakfast consisted of porridge and milk, his dinner bread and beer, and porridge and milk again for supper. A Chinaman would starve upon such food (1857:42–43, cited by Anderson 1988:96).

Fortune was observing mid-nineteenth-century China, a land scholars generally view as already subject to a growing crisis. How much truer his assessment may have been for the century preceding. No wonder that at least one modern specialist has asserted confidently that "the Chinese peasant of the Yongzheng [1723–1735] and the first half of the Qianlong [1736–1765] eras was in general better nourished and more comfortable than his French counterpart during the reign of Louis XV."[10] In short, the fragmentary evidence on nutrition and living standards offers little ground for concluding that Chinese standards of living were so low as to make plausible higher mortality rates because of resource limitations. Some foreign observers may have been reacting to food supply issues in cultural terms of taste and convention more than scientific terms of nutritional quality.

Nevertheless, there is no doubt that late imperial China was subject to famine. Particularly in the nineteenth century, crises created by some combination of natural and human factors raised mortality across different regions of China. Beginning at mid-century, rebellions hit the Yangzi River valley and the Huai River to its north, as well as parts of Northwest, Southwest, and South China. The worst natural disasters occurred in North and Northwest China, where years of successive drought between 1876 and 1879 depleted grain reserves and drove peasants to search for food. Where drought failed to wither the crops, armies interrupted the agricultural cycle and extracted much of what was left. We can expect that mortality rose as increased malnutrition made people more vulnerable to disease and as some succumbed to starvation.

Foreign and domestic observers alike recorded their bleak assessments of this period. For Qingzhou in the North China province of Shandong the Dutch minister J. H. Ferguson estimated between 30 and 60 percent of the families in many villages had been wiped out by famine, while the English Baptist missionary Timothy Richard reported a death rate reaching 90 percent in some smaller villages (Bohr 1972:15). To Shandong's west, Shanxi's governor, Zeng Guochuan, reported in late 1879 that some 80 percent of the population had been affected by the recent famine, with 60 to 70 percent suffering from typhoid fever (Bohr 1972:23). Bohr (26) estimates some nine and a half million people may have died in the North and Northwest.

[10]Gernet 1972:420–21. The English translation erroneously has "in general, much better and much happier" (1982:481).

As grim as these accounts are, it is uncertain that such crises set China apart from Europe. Early modern Europe also experienced sharp mortality peaks resulting from famines, epidemics, and wars. The plague was a major killer in the fourteenth and fifteenth centuries: wars were less devastating directly but because they interrupted production and trade could lead to subsistence crises (Hohenberg and Lees 1985:79–83). Scholars have discovered that in the aftermath of some mortality crises, fertility rises to accelerate the replacement of lost populations. Thus mortality crises appear to have at most a temporary impact on population and resource issues (Flinn 1981:25–47); Charbonneau and LaRose 1979; Bongaarts and Cain 1980). Subsistence crises continued to be a problem even in nineteenth-century Europe. Such crises were still a serious threat for some populations early in the century, and they continued to cause fear even at mid-century (Post 1976; McPhee 1992:57–61). These problems did not block industrialization. It seems unlikely therefore that population and resource issues, at least as reflected in mortality crises, limited the possibilities of industrialization.

Whatever the impact of crisis mortality on population dynamics, a more basic similarity between Chinese and European mortality patterns is obscured: overall, life expectancy was the same in Europe and in China. This similarity contradicts a growing body of data indicating that Chinese mortality levels rose continuously from the late seventeenth century into the nineteenth century. Ted Telford, for example, argued on the basis of genealogical data for Tongcheng county in Anhui that life expectancies at birth dropped from 39.6 between 1750 and 1769 to 33.4 a half-century later (Telford 1990:133). Liu Ts'ui-jung (1985, 1992) also found dramatic declines in life expectancy during the eighteenth century. There is reason to doubt, however, that such a sharp downward trend in life expectancy occurred. Even the genealogical demographers themselves note problems with their data (Harrell 1987; Telford 1990). As the genealogies become more inclusive of infants and children, for example, as they do over time, mortality rates seem to rise. The parallel declines in life expectancy found by Liu and Telford may be artifacts of changes in the way the genealogies were recorded.[11] Nor do the reported trends match the historical context. It is perplexing to find life expectancy highest during the economic recession of the late seventeenth century, when officials, especially those along the Yangzi River, feared for the poor peasants who received low prices for their crops and for the laborers who could not find employment (Kishimoto 1984). As crop prices rose and employment grew throughout the eighteenth century. Telford reports rising mortality. Of course, the very low levels of life expectancy attained in the mid-nineteenth century could have been influenced by the Taiping Rebellion, but this would have been a temporary phenomenon.

Probably, therefore, life expectancy in China did not decline throughout the eighteenth and nineteenth centuries as the genealogical data suggest. But even if we assume these data are correct, the overall similarity of Chinese and European life expectancies remains. Except in the most prosperous regions of Europe, life expectancy did not surpass the Chinese level until the end of the nineteenth century (see Table 1).[12] Thus it seems quite unlikely that resource shortages affected mortality in China but not in Europe. Malthus's argument that eighteenth-century European populations were freed from positive checks while Chinese populations were not seems to be untenable.

Economic Dynamics and Preconditions of Capitalism

But what of the potential for Smithian growth in China? Several works on China's agrarian economy depict China more in Malthusian terms, seeing in it a dynamic equilibrium between population growth and economic expansion that fits within the classical economists's world of limited possibilities. Most explicitly in discussing the population–resource issue was Dwight Perkins's 1969 study of Chinese agriculture from the fourteenth century forward. Perkins assumed that per capita grain consumption remained roughly constant over a six-hundred-year period, an assumption based on his estimates of rising land productivity, extended arable, and total population. He then addressed various other factors responsible for the continued balance of population and resources—seeds, cropping patterns, new crops, farm implements, water control, fertilizer, and grain marketing.

Ping-ti Ho's classic study of Chinese population also addressed the problem of population and resources in Malthusian terms (Ho 1959). In Ho's opinion, the nineteenth-century rebellions and natural disasters were Malthusian checks on a population that had reached an optimal level in the late eighteenth century. Ho argued that China's population and resources were forced into equilibrium through checks on population growth rather than by a continuing expansion of the resource base. Ho's analysis of these issues is the only major work that explicitly appeals to Malthus's writings, but not the only one to fit within the framework created by European classical economists. Kang Chao (1986), for example, offers a sweeping view of Chinese agricultural history in which Chinese demographic growth matches any and all increases in aggregate production, canceling out any per capita economic growth. Chao suggests that cultural values led Chinese people to have large families. Like the classical economists, Chao describes a situation in which an equilibrium between population and resources brings people close to subsistence.

Even works that are not obviously within the framework of classical economics offer arguments and findings that affirm an equilibrium between population

[11]Telford attributes the early high life expectancies in the Toncheng lineage to this factor. He rejects the estimates of life expectancy at birth (E_0) of over 40 years before 1750, and calls 1750–1769 male life expectancy of 39.6 the likely peak of life expectancy during the Qing (1990:133). However, there is no evidence that reporting of infants and children was complete by 1750. A gradual improvement in completeness would explain why, over a series of cohorts from 1790–1809 to 1860–1879, infant and child mortality rises monotonically, while adult mortality remains basically unchanged (1990:fig. 8).

[12]These issues are discussed at greater length in Lavely and Wong 1991.

and European populations

	Period	Life Expectancy	Source
German villages[a]	Pre–1800	35.1	Knodel 1988:59
	1800–1849	38.7	
	1850+	39.4	
Geneva	1600–1649	32.2	Henry 1956:158
	1650–1699	31.9	
	1700–1749	41.6	
	1750–1799	47.3	
	1800–1849	51.6	
	1850–1899	57.9	
British peeragee (males)	1330–1479	24.0	Hollingsworth 1965:358
	1480–1679	27.0	
	1680–1729	33.0	
	1730–1779	44.8	
	1780–1829	47.8	
	1830–1879	49.8	
	1850–1954	54.6	
French Villages[b]	Pre-1750	25.0	Flynn 1981:130–31
	1740–1790	30.0	
	1780–1820	36.0	
Tongcheng lineages (males)	1690–1709	46.2	Telford 1990:133
	1710–1729	42.1	
	1730–1749	40.7	
	1750–1769	39.6	
	1770–1789	38.2	
	1790–1809	33.4	
	1800–1819	34.9	
	1820–1839	31.1	
	1840–1859	28.2	
	1860–1879	24.5	
Zhejiang Shen[c]	1725–1739	36.0	Liu 1985:52
	1740–1754	38.5	
	1755–1769	38.5	
	1770–1784	38.5	
	1785–1799	36.0	
	1800–1814	36.0	
	1815–1829	31.2	
	1830–1844	31.2	
Liaoning villages[d]	1792–1867	35.9	Lee and Campbell 1997:62

[a]According to Coale-Demeny (1966) model west.

[b]Model west female E(0) implied by mean 1(10).

[c]Males, period rates.

[d]Males, ages 1–5 *sui*; *sui* indicates the number of calendar years during which a person has lived. People are one *sui* at birth and two *sui* at the next New Year.

and resources. Philip Huang has argued, first for northern Chinese agriculture, and then for the more productive region of the lower Yangzi, that Chinese cases lacked the kind of market development that led England to industrial capitalism (Huang 1985, 1990). The assumption of European market expansion leading to industrial capitalism misses the economic and demographic similarities of China and Europe before the Industrial Revolution. When Huang argues that population and resources remained in a rough balance to create a persisting subsistence economy in China, he unintentionally echoes assessments made of early modern Europe. Mark Elvin (1973) shares Huang's assumption that market expansion should lead to broader economic changes, but unlike him, Elvin stresses the dramatic ways in which China's commercial economy developed in the Song dynasty and expanded in the Ming and Qing dynasties. Elvin puzzles over why these changes ended in a "high-level equilibrium trap." The mystery is considerably reduced if we recognize that Elvin's assessment of late imperial China's market economy fits within a Smithian framework. This Smithian framework need not mean an end to the Malthusian limitations of population and resources. The view of agrarian China as a massive society in which population and resources broadly stayed in a balance over many centuries remains persuasive, but this balance may have been considerably above a subsistence minimum. James Lee and his associates, for example, that have shown the Chinese demographic system was far more regulated than we previously believed. On the economic side, Smithian dynamics were responsible for much of the economic growth that did take place. As commercial exchange fostered the development of more sophisticated merchant organizations and stimulated market activity, common people in both China and Europe were drawn into market exchange. In China, market demand induced additional investments in water control, the search for higher and more stable yield seed varieties, and shifts to alternative cash crops.

But Smithian dynamics need not necessarily lead to sustained increases in per capita incomes. Outside China, in the heart of Europe's developing capitalism, including England during the centuries of early modern growth, real wages over the long run did not change. The dramatic change in real wages and in the ratio of population to resources did not come until nineteenth-century urban factory industrialization (Levine 1987). Over the long run, pre-nineteenth-century populations generally adjusted to economic conditions. In the short run, however, there were all manner of fluctuations driven by fertility and mortality responses; wages could fall as well as rise. Chinese population and resources probably stayed approximately in balance, with cycles of rising and falling standards of living that remained widely variable across regions over many centuries. Although no radical rupture permitting sustained per capita growth took place, few places have experienced such a break endogenously.

More important perhaps is to explain how the lower Yangzi achieved the considerable economic success that it did. This area was one of Eurasia's most consistently dynamic economies between 1350 and 1750. Chinese commercialization clearly helped make the "long run" in which it became possible for increasing numbers of people to survive on ever-smaller allotments of land.

Jan de Vries (1993, 1994) characterized some of the changes taking place in seventeenth and eighteenth-century Europe as an "industrious revolution." Between 1600 and 1800 increasing numbers of Europeans, including peasants, became dependent upon market purchases. This demand stimulated market expansion at a time when per capita incomes were not rising, and indeed may have even been falling. This seems counterintuitive. Why did people buy and sell more goods if their wages did not increase? De Vries explains that Europeans began to work harder; they became more "industrious." As evidence he cites the reduction of nonworking days for both artisans and peasants, and he suggests that people became more willing to trade leisure for money. Thus de Vries's argument explains the expansion of early modern Europe's commercial economy without requiring rising per capital incomes.

De Vries argues for an "industrious revolution" in order to shift our attention from the supply-side innovations of the Industrial Revolution to an earlier set of demand-side shifts. But is this industrious revolution simply a European phenomenon?[13] The Smithian dynamics of growth found in China share at least some features of the industrious revolution de Vries describes. Increased labor for commercial production also characterized parts of China between the sixteenth and eighteenth centuries. This increase in market-oriented labor produced thriving silk and cotton craft industries in the lower Yangzi region; women's increased labor for market production was especially crucial in the expansion of textiles (Huang 1990:44–57). In agriculture, new crop rotations and clearance of additional land reduced the number of days in the slack season as peasants aimed to produce additional crops for market sale. Labor intensification accompanied Smithian growth in China as it did in Europe.

Philip Huang calls this process of increased market production in which Chinese peasants worked harder "involution," because returns on each workday appear to decline as laborers increase the number of days they work annually. For Huang involution is the centerpiece of how Chinese economic growth differs from European economic development. But how different is Huang's lower Yangzi from de Vries's Dutch Republic and England? In Europe too people worked harder without necessarily experiencing rising per capita incomes. Such similarities between early modern Europe's industrious revolution and what Huang labels "growth without development" in China deserve further analysis. For the moment it appears that de Vries's industrious revolution and Huang's involution share some important features, and these features were part of the Smithian dynamics of market-based growth supported by labor intensification in the advanced regions of China and in Europe in the centuries preceding the Industrial Revolution.

The parallels between China and Europe deserve emphasis because they are so deeply obscured in the best analyses of Chinese and European patterns of economic change. Certain caveats, however, are in order. First, more research is needed. Without data on Chinese patterns of consumption in the late imperial period, for example, we cannot determine whether the economic impacts of Chinese and European consumption were fully parallel. Second, the *degrees* to which peasants in various settings can intensify their labor and reduce their slack season unemployment differ. One of the basic issues in twentieth-century peasant societies has been the recalcitrant problem of surplus agricultural labor. Problems that afflicted China in the 1920s and 1930s, however, may not have existed in the 1720s and 1730s. In the earlier period, peasants in some regions may have found constructive ways to work harder more easily: fewer people were searching for ways to be productively employed, and they could exploit techniques that were no longer new in the twentieth century. The industrious revolution that could not occur now, therefore, may have occurred then. Third, the supply-driven Industrial Revolution that followed the demand-led industrious revolution in Europe did not happen in China. De Vries argues for an unbreakable link between the changing consumption patterns of early modern Europe and the Industrial Revolution, but it is by no means clear that such a link is inevitable. We have seen that China too manifested certain Smithian patterns of economic growth, as well as areas where intensified labor did not raise per capita incomes, so these factors alone cannot explain the European Industrial Revolution of the nineteenth century. In studying what set China apart, we may acquire a better understanding of both Chinese and European developments.

Conclusion

Both the early modern English and the late imperial Chinese agrarian economies were subject to the positive and negative forces of change discussed by Adam Smith and Thomas Malthus. This is not to say that England and the lower Yangzi manifested exactly the same economic and demographic behaviors, for certainly there were differences. Some of these differences, however, turn out to have been more apparent than real.[14] China and Western Europe shared a common world

[13]De Vries (1993:126) acknowledges a prior use of the term "industrious revolution" to describe changes in nineteenth-century Japanese economic history by Akira Hayami. Hayami (1989) proposed the term as an alternative to Europe's "Industrial Revolution" to describe how Japan and Europe differed. Japan, in this argument, relied more on intensification of labor than on capital to achieve growth.

[14]For instance, Philip Huang makes much of the many restrictions on the competitive functioning of labor and credit markets (1990:106–11). He implicitly measures Chinese markets against the ideals taught in a first-year economics course. In the real world of early modern Europe, constraints on markets were significant too. Indeed, rural land and credit markets were often far less free than those in China that Huang finds so limited (Goody, Thirsk, and Thompson 1976:25, 126–27, 132–33; Hoffman, Postel-Vinay, and Rosenthal 1992; Rosenthal 1993). European and Chinese markets both diverged from theoretical ideals; the greater challenge is to explain how and where the dynamics in the two cases run parallel and then diverge.

of harvest insecurities and material limitations. Both suffered through cycles of economic expansion and contraction that created gradually larger economies propelled forward by similar Smithian dynamics of spatial divisions of labor and comparative advantage through the market.

China and Europe shared the demographic possibilities sketched out by Malthus. Contrary to stereotypes, before 1800 China was not clearly more vulnerable than Europe to economic and demographic crisis. Even if future scholarship affirms important differences in demographic rates in China and Europe, the presence of a potentially fragile population–resource ratio will remain common to both ends of Eurasia. Even if arguments about a considerable margin of prosperity in early modern Europe demonstrate that a Malthusian danger was not yet near at hand, Smithian dynamics alone could not guarantee that such a danger would remain distant. The economic cores of both China and Europe may have sustained a considerable margin as true Malthusian pressures weighed down on China's ecologically more precarious peripheries by the mid-nineteenth century. Whatever future research suggests with respect to these large issues, the limits of economic possibility within a world defined by classical economists remain relevant for both China and Europe (and the rest of the world) before the nineteenth century. When we turn in the next chapter to industry in China and Europe during the seventeenth and eighteenth centuries, we once again will discover surprising similarities, further narrowing the terrain on which we can expect to locate crucial economic differences.

2

DYNAMICS OF INDUSTRIAL EXPANSION IN EARLY MODERN EUROPE AND LATE IMPERIAL CHINA

Though European industrial production never accounted for a large portion of economic activity before 1800, the dynamics of industrial expansion occupy a privileged place in historiography because of what industry would come to be in the nineteenth and twentieth centuries. Economic historians study industrialization retrospectively in search of the paths leading to modern economies. In particular, the precursors to the Industrial Revolution that burst forth in England in the late eighteenth century are assumed to be more general factors that could promote similar economic change elsewhere. We can identify three periods of European industrial activity beginning in the early modern period: (1) late fifteenth to early sixteenth centuries: urban craft production; (2) mid-sixteenth to mid-eighteenth centuries: rural cottage industry; (3) late eighteenth to late nineteenth centuries: urban factory mechanization. The shift from urban production controlled by craft guilds to rural cottage industry is generally viewed as the liberation of production from the restrictions imposed to protect producer interests; this becomes one sign of the breakdown of "feudal" control. The subsequent shift of industry back into the cities is usually seen as the triumph of the Industrial Revolution, which heralds the ascendancy of the bourgeoisie. In between lies the period of rural industry when neither "feudal" nor "capitalist" elements were clearly in control in Europe.[1]

[1] I have expressed the changes within a Marxist set of categories because such a formulation highlights the framework within which many Chinese and Japanese economic historians look at Chinese and European industrialization. But even without the use of "feudalism" and "capitalism" as basic categories, the phenomenon of rural industry is widely recognized as an important precursor to the Industrial Revolution.